

Table 1: WikiText-2 test perplexity of FP16, BFP, BiE, and activation-only BiE (ABiE). Lower is better.

| Method | LLaMA-2-7B | LLaMA-3.1-8B | OPT-6.7B |
|--------|------------|--------------|----------|
| FP16   | 5.47       | 6.24         | 10.86    |
| BFP6   | 5.52       | 6.33         | 10.89    |
| BiE6   | 5.49       | 6.29         | 10.89    |
| ABiE6  | 5.51       | 6.30         | 10.89    |
| BFP5   | 5.63       | 6.62         | 11.14    |
| BiE5   | 5.57       | 6.44         | 10.97    |
| ABiE5  | 5.60       | 6.52         | 11.12    |
| BFP4   | 6.12       | 7.77         | 13.02    |
| BiE4   | 5.86       | 7.06         | 12.64    |
| ABiE4  | 5.96       | 7.34         | 12.74    |